Danica Xiong

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Education

Stanford University — *Stanford, CA*
M.S. in Computer Science — *Sept 2023 – June 2025*
- 3.97 GPA: Specialization in Visual Computing (Computer Vision, Graphics, Robotics)

University of California, San Diego — *San Diego, CA*
B.S. in Computer Science and Engineering — *Sept 2018 – March 2023*
- 3.92 GPA: Magna Cum Laude Latin honors, Warren Honor Society, 8x Provost Honors Recipient
- 4.0 GPA: Minor in Mathematics

Experience

Nvidia — *Santa Clara, CA*
System Software Engineer IC2 — *Sept 2025 – Present*
- Worked concurrently on 2 teams: GPU Simulation (sim chip output) & LLM tooling (Agents & RAG)
- Proposed & implemented architectural changes to combine Compute Warp States and Mesh Thread Groups into a unified compute shader class through Bridging. Refactoring removed the need for devs to implement compute functions twice due to virtual inheritance structure
- Built knob validation system, on p4 submission, checks if knobs have been categorized and warns user to remove from deprecation list
- On LLM team, shipped: Keyword filtering, multi file & link indexing & reading, streaming delay, database migrations, terminal agents, agent configurations and tools, agent JSON/PNG/HTML/LOG parsing, hybrid RAG with Glean, Agents for Perforce, regression dashboards, bug fixes
- Re-orged on to AI Tooling team and shipped: SASS assembly behaviour diff tool, automatic ISA to Op code library workflow, GPU Simulation performance auditing, hardware constraint verification, server rack allocation optimization, EDA efficiency anomoly detection.
- Worked with hardware developers on 8+ teams (including formal verification, chip sim, perf sim, SASS, MODS, ISA, Server farm, electronic design etc) to identify areas that need tools, define requirements, build tools, tested and iterated until tools solved use cases
- Completely automated ISA to Op code development workflow with handoff to continue automation into GPU modeling. 100% accuracy over 5 months in production shipping 100+ change lists.

Corpia — *San Francisco, CA*
Cofounder - Chief Technology Officer — *April 2026 – June 2026*
- Built enterprise context layer + team agent context from agent traces. Accepted YCombinator S26, company evaluated at 40 Million
- Presented to VP of Qualcomm, VP of Merck, 5 midsized companies, acquired 4 startup customers, conducted customer discovery meetings
- Designed and built entire context layer pipeline, deployed on AWS, database setup, w/ multi-tenancy architecture, iterated on w/ customers
- Improved retrieval latency, experimented with synthesis layers and agent memory structures, got SOTA on Evolving Memory Evaluation

OpenAI — *San Francisco, CA*
Member of Technical Staff — *July 2025 - Sept 2025*
- Designed, built, shipped, and maintained 3 new products: ChatGPT API, Apple Intelligence & Kakao, worked with legal, UI, partner companies
- Built new security scoping restricting 3rd party access in Rust & go. Migrated codebase, async workers, synthetics to comply with new auth
- Integrated biscuit auth between services to mitigate abuse. Developed unit and integration tests to test across every service owned by team
- Debugged image gen and file gen pipelines, debugged synthetics, identified bugs via Datadog to minimize/prevent tool downtime in Python

Nvidia — *Santa Clara, CA*
System Software Engineer Intern — *Sept 2024 – Dec 2024*
- Built and deployed new full stack portal for devs to upload and share API documentation across teams, complying w/ Microsoft Azure auth
- Built and deployed CI/CD pipeline for front and backend, set up infrastructure to automatically generate docker image and host internally
- Researched and wrote plan for incremental dependency upgrades, API efficiency and code improvements, identified and fixed API edge cases
- Integrated experimental internal LLM Nvidia Nemo as a chatbot with chip scheduling client: Automatically answered user questions
- Wrote automatic API documentation generation script using internal LLM fine-tuned on company confluence pages, removing need to write documentation, reducing development lifecycle from 3 weeks to 5 minutes, removing 10+ year old dependencies on CRNK and Springfox

Riot Games — *Los Angeles, CA*
Software Development Intern — *June 2024 – Sept 2024*
- Designed and implemented a dynamic map system in the LoL game engine, enabling projection of 3D geometry onto a 2D map
- Developed and optimized graphics engine components: vertex & frag shaders, UV mapping, and polygon convexity & handedness issues
- Parameterized logic and graphics engine calls, creating internal tool blocks to allow designers to use new features without needing to code
- Updated game client & data warehouse to ingest new telemetry data, enabling the collection of time-based user inventory statistics
- Optimized server performance by moving shape calculation to the client side, reducing network traffic from 1000+ packets to one

Stanford Computer Graphics Lab — *Stanford, CA*
Research Assistant — *Oct 2023 – Mar 2024*
- Built low latency video platform to simulate professional conference spaces and physical interactions, approved by Denis Zorin
- Integrated Zoom Video SDK APIs, low-level video developer APIs, and data pipeline to dynamically manage conference attendees in real-time
- Built proof of concept for 250+ worldwide SIGGRAPH 2024 committee members to collaborate on technical reviews

Center for Visual Computing — *San Diego, CA*
Undergraduate Researcher — *Sep 2022 – Mar 2023*
- Built Python Compiler Javana that compiles inverse rendering programs to assembly to optimize for register mapping and spilling
- Benchmarked optimal automatic differentiation (Forward vs Backward) workflow for differential rendering, sped up performance by 100%

Amazon — *Austin, TX*
Software Development Engineer Intern — *Jun 2022 – Sep 2022*
- Developed data visualization portal that compares time-series ML model forecasted sales against customer grocery sales for supply chain
- Created and revised database and application designs, approved by Senior Software Developers, Managers, and Amazon Scientists
- Cleaned and optimized data using AWS services (S3, Lambda, Dynamo, Aurora, API Gateway) for architecting a new data pipeline
- Ensured compliance with Amazon's infrastructure standards, including CORS, token verification, and role-based access control
- Deployed across US, EU, and IN for internal backtesting and external customers visualization

Awards & Publications

2024	**YCombinator Spring 2024 Batch & Best Paper**, of 263 papers CS231N (FeiFei Li's class) HiLait: Automatic Video Highlighting System Leveraging Audio, Text, Facial, and Semantic AI	*Stanford, CA*
2024	**Champion**, of 78 teams. Best reasoning, search & representation algorithms in CS227B General Gameplay	*Stanford, CA*
2023	**First Place**, Built math-based Non-Euclidean shader displaying hyperbolic and elliptic space	*VC Hackathon*
2022	**Co-Author**, UnMapped: Exploiting Experts' Situated Experiences in Collaborative Mixed Reality	*SIGCHI 2023*
2022	**Presenter**, Developed and demoed real time 3D networked engine at Qualcomm Institute, that featured realtime path tracing with hardware supported acceleration	*San Diego, CA*

Skills

Languages	Java, C++, C, Python, Haskell, GLSL, HLSL, Javascript, Typescript, LaTeX, Rust, System Verilog, C#, Julia
Artificial Intelligence/ML	Pytorch, OpenCV, Numpy, Pandas, Matplotlib, CVXPY
Graphics/Game Dev	DirectX11, OpenGL, Unity, Steam VR, Oculus, Autodesk Maya, DirectX12, DirectX9
Web Development	React, HTML, CSS, Rest API, Node, JWT Tokens, Spring, MongoDB, Tomcat, VueJS, Docker, SQL, Maven
Cloud & AWS Technology	S3, Lambda, DynamoDB, AuroraDB, Athena, Glue, EC2, CloudFormation, CloudWatch
Software tools	VI, Perforce, Github, Git, Gitlab, Github Actions, Agile, GDB, Valgrind, Bash